

Complex Numbers and Functions. Complex Differentiation

The transition from “real calculus” to “complex calculus” starts with a discussion of *complex numbers* and their geometric representation in the *complex plane*. We then progress to *analytic functions* in Sec. 13.3. We desire functions to be analytic because these are the “useful functions” in the sense that they are differentiable in some domain and operations of complex analysis can be applied to them. The most important equations are therefore the Cauchy–Riemann equations in Sec. 13.4 because they allow a test of analyticity of such functions. Moreover, we show how the Cauchy–Riemann equations are related to the important *Laplace equation*.

The remaining sections of the chapter are devoted to elementary complex functions (exponential, trigonometric, hyperbolic, and logarithmic functions). These generalize the familiar real functions of calculus. Detailed knowledge of them is an absolute necessity in practical work, just as that of their real counterparts is in calculus.

Complex Numbers and Their Geometric Representation

The material in this section will most likely be familiar to the student and serve as a review.

Equations without *real* solutions, such as $x^2 = -1$ or $x^2 - 10x + 40 = 0$, were observed early in history and led to the introduction of complex numbers.¹ By definition, a **complex number** z is an ordered pair (x, y) of real numbers x and y , written

$$z = (x, y).$$

x is called the **real part** and y the **imaginary part** of z , written

$$x = \operatorname{Re} z, \quad y = \operatorname{Im} z.$$

By definition, two complex numbers are **equal** if and only if their real parts are equal and their imaginary parts are equal.

$(0, 1)$ is called the **imaginary unit** and is denoted by i ,

$$(1) \quad i = (0, 1).$$

Addition, Multiplication. Notation $z = x + iy$

Addition of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is defined by

$$(2) \quad z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

Multiplication is defined by

$$(3) \quad z_1 z_2 = (x_1, y_1)(x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1).$$

These two definitions imply that

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0)$$

and

$$(x_1, 0)(x_2, 0) = (x_1 x_2, 0)$$

as for real numbers x_1, x_2 . Hence the complex numbers “*extend*” the real numbers. We can thus write

$$(x, 0) = x. \quad \text{Similarly,} \quad (0, y) = iy$$

because by (1), and the definition of multiplication, we have

$$iy = (0, 1)y = (0, 1)(y, 0) = (0 \cdot y - 1 \cdot 0, 0 \cdot 0 + 1 \cdot y) = (0, y).$$

Together we have, by addition, $(x, y) = (x, 0) + (0, y) = x + iy$.

In practice, complex numbers $z = (x, y)$ are written

$$(4) \quad z = x + iy$$

or $z = x + yi$, e.g., $17 + 4i$ (instead of $i4$).

Electrical engineers often write j instead of i because they need i for the current.

If $x = 0$, then $z = iy$ and is called **pure imaginary**. Also, (1) and (3) give

$$(5) \quad i^2 = -1$$

because, by the definition of multiplication, $i^2 = ii = (0, 1)(0, 1) = (-1, 0) = -1$.

For **addition** the standard notation (4) gives see (2)

$$(x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2).$$

For **multiplication** the standard notation gives the following very simple recipe. Multiply each term by each other term and use $i^2 = -1$ when it occurs see (3) :

$$\begin{aligned}(x_1 + iy_1)(x_2 + iy_2) &= x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2 \\ &= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1).\end{aligned}$$

This agrees with (3). And it shows that $x + iy$ is a more practical notation for complex numbers than (x, y) .

If you know vectors, you see that (2) is vector addition, whereas the multiplication (3) has no counterpart in the usual vector algebra.

EXAMPLE Real Part, Imaginary Part, Sum and Product of Complex Numbers

Let $z_1 = 8 + 3i$ and $z_2 = 9 - 2i$. Then $\operatorname{Re} z_1 = 8$, $\operatorname{Im} z_1 = 3$, $\operatorname{Re} z_2 = 9$, $\operatorname{Im} z_2 = -2$ and

$$\begin{aligned}z_1 + z_2 &= (8 + 3i) + (9 - 2i) = 17 + i, \\ z_1z_2 &= (8 + 3i)(9 - 2i) = 72 + 6 + i(-16 + 27) = 78 + 11i.\end{aligned}$$

Subtraction, Division

Subtraction and **division** are defined as the inverse operations of addition and multiplication, respectively. Thus the **difference** $z = z_1 - z_2$ is the complex number z for which $z_1 = z + z_2$. Hence by (2),

$$(6) \quad z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2).$$

The **quotient** $z = z_1/z_2$ ($z_2 \neq 0$) is the complex number z for which $z_1 = zz_2$. If we equate the real and the imaginary parts on both sides of this equation, setting $z = x + iy$, we obtain $x_1 = x_2x - y_2y$, $y_1 = y_2x + x_2y$. The solution is

$$(7) \quad z = \frac{z_1}{z_2} = x + iy, \quad x = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2}, \quad y = \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}.$$

The **practical rule** used to get this is by multiplying numerator and denominator of z_1/z_2 by $x_2 - iy_2$ and simplifying:

$$(7) \quad z = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}.$$

EXAMPLE 2 Difference and Quotient of Complex Numbers

For $z_1 = 8 + 3i$ and $z_2 = 9 - 2i$ we get $z_1 - z_2 = (8 + 3i) - (9 - 2i) = -1 + 5i$ and

$$\frac{z_1}{z_2} = \frac{8 + 3i}{9 - 2i} = \frac{(8 + 3i)(9 + 2i)}{(9 - 2i)(9 + 2i)} = \frac{66 + 43i}{81 + 4} = \frac{66}{85} + \frac{43}{85}i.$$

Check the division by multiplication to get $8 + 3i$.

Complex numbers satisfy the same commutative, associative, and distributive laws as real numbers (see the problem set).

Complex Plane

So far we discussed the algebraic manipulation of complex numbers. Consider the geometric representation of complex numbers, which is of great practical importance. We choose two perpendicular coordinate axes, the horizontal x -axis, called the **real axis**, and the vertical y -axis, called the **imaginary axis**. On both axes we choose the same unit of length (Fig. 318). This is called a **Cartesian coordinate system**.

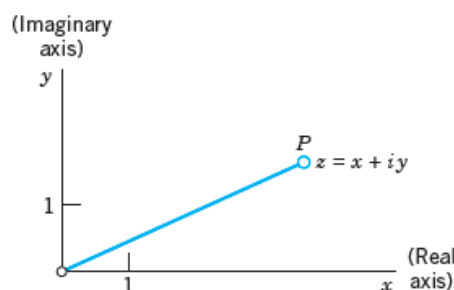


Fig. 318. The complex plane

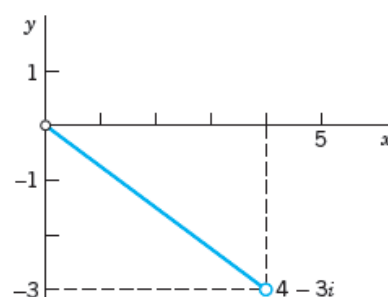


Fig. 319. The number $4 - 3i$ in the complex plane

We now plot a given complex number $z = (x, y) = x + iy$ as the point P with coordinates x, y . The xy -plane in which the complex numbers are represented in this way is called the **complex plane**.² Figure 319 shows an example.

Instead of saying “the point represented by z in the complex plane” we say briefly and simply “*the point z in the complex plane*.” This will cause no misunderstanding.

Addition and subtraction can now be visualized as illustrated in Figs. 320 and 321.

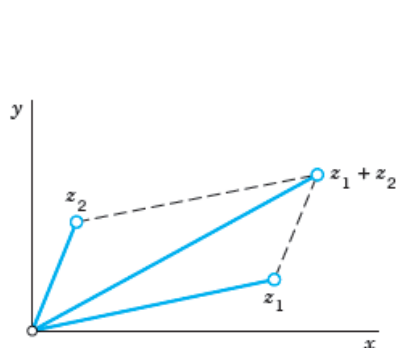


Fig. 320. Addition of complex numbers

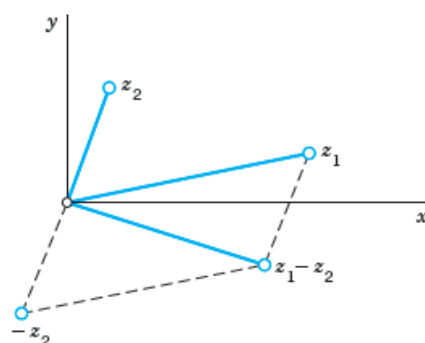


Fig. 321. Subtraction of complex numbers

Complex Conjugate Numbers

The **complex conjugate** $\bar{z}$ of a complex number $z = x + iy$ is defined by

$$\bar{z} = x - iy.$$

It is obtained geometrically by reflecting the point z in the real axis. Figure 322 shows this for $z = 5 + 2i$ and its conjugate $\bar{z} = 5 - 2i$.

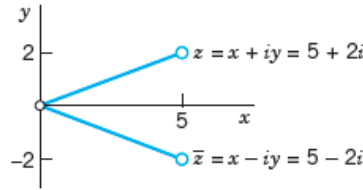


Fig. 322. Complex conjugate numbers

The complex conjugate is important because it permits us to switch from complex to real. Indeed, by multiplication, $z\bar{z} = x^2 + y^2$ (verify). By addition and subtraction, $z + \bar{z} = 2x$, $z - \bar{z} = 2iy$. We thus obtain for the real part x and the imaginary part y (not iy) of $z = x + iy$ the important formulas

$$(8) \quad \operatorname{Re} z = x = \frac{1}{2}(z + \bar{z}), \quad \operatorname{Im} z = y = \frac{1}{2i}(z - \bar{z}).$$

If z is real, $z = x$, then $\bar{z} = z$ by the definition of $\bar{z}$, and conversely. Working with conjugates is easy, since we have

$$(9) \quad \begin{aligned} \overline{(z_1 + z_2)} &= \bar{z}_1 + \bar{z}_2, & \overline{(z_1 - z_2)} &= \bar{z}_1 - \bar{z}_2, \\ \overline{(z_1 z_2)} &= \bar{z}_1 \bar{z}_2, & \overline{\left(\frac{z_1}{z_2}\right)} &= \frac{\bar{z}_1}{\bar{z}_2}. \end{aligned}$$

EXAMPLE 3 Illustration of (8) and (9)

Let $z_1 = 4 + 3i$ and $z_2 = 2 + 5i$. Then by (8),

$$\operatorname{Im} z_1 = \frac{1}{2i}(4 + 3i) - (4 - 3i) = \frac{3i + 3i}{2i} = 3.$$

Also, the multiplication formula in (9) is verified by

$$\begin{aligned} \overline{(z_1 z_2)} &= \overline{(4 + 3i)(2 + 5i)} = \overline{(-7 + 26i)} = -7 - 26i, \\ \bar{z}_1 \bar{z}_2 &= (4 - 3i)(2 - 5i) = -7 - 26i. \end{aligned}$$

13.2 Polar Form of Complex Numbers. Powers and Roots

We gain further insight into the arithmetic operations of complex numbers if, in addition to the xy -coordinates in the complex plane, we also employ the usual polar coordinates r, θ defined by

$$(1) \quad x = r \cos \theta, \quad y = r \sin \theta.$$

We see that then $z = x + iy$ takes the so-called **polar form**

$$(2) \quad z = r(\cos \theta + i \sin \theta).$$

r is called the **absolute value** or **modulus** of z and is denoted by $|z|$. Hence

$$(3) \quad |z| = r = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}.$$

Geometrically, $|z|$ is the distance of the point z from the origin (Fig. 323). Similarly, $|z_1 - z_2|$ is the distance between z_1 and z_2 (Fig. 324).

θ is called the **argument** of z and is denoted by $\arg z$. Thus $\theta = \arg z$ and (Fig. 323)

$$(4) \quad \tan \theta = \frac{y}{x} \quad (z \neq 0).$$

Geometrically, θ is the directed angle from the positive x -axis to $\mathbf{OP}$ in Fig. 323. Here, as in calculus, all *angles are measured in radians and positive in the counterclockwise sense*.

For $z = 0$ this angle θ is undefined. (Why?) For a given $z \neq 0$ it is determined only up to integer multiples of 2π since cosine and sine are periodic with period 2π . But one often wants to specify a unique value of $\arg z$ of a given $z \neq 0$. For this reason one defines the **principal value** $\text{Arg } z$ (with capital A) of z by the double inequality

$$(5) \quad -\pi < \text{Arg } z \leq \pi.$$

Then we have $\text{Arg } z = 0$ for positive real $z = x$, which is practical, and $\text{Arg } z = \pi$ (not $-\pi$) for negative real z , e.g., for $z = -4$. The principal value (5) will be important in connection with roots, the complex logarithm (Sec. 13.7), and certain integrals. Obviously, for a given $z \neq 0$, the other values of $\arg z$ are $\arg z = \text{Arg } z \pm 2n\pi$ ($n = \pm 1, \pm 2, \dots$).

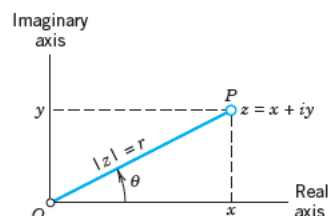


Fig. 323. Complex plane, polar form of a complex number

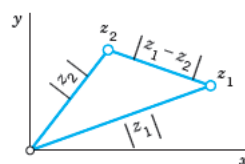


Fig. 324. Distance between two points in the complex plane

EXAMPLE 1

Polar Form of Complex Numbers. Principal Value $\text{Arg } z$

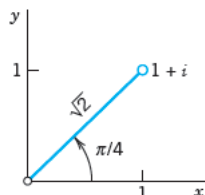


Fig. 325. Example 1

$z = 1 + i$ (Fig. 325) has the polar form $z = \sqrt{2} (\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi)$. Hence we obtain

$$|z| = \sqrt{2}, \quad \arg z = \frac{1}{4}\pi \pm 2n\pi \quad (n = 0, 1, \dots), \quad \text{and} \quad \text{Arg } z = \frac{1}{4}\pi \quad (\text{the principal value}).$$

Similarly, $z = 3 + 3\sqrt{3}i = 6 (\cos \frac{1}{3}\pi + i \sin \frac{1}{3}\pi)$, $|z| = 6$, and $\text{Arg } z = \frac{1}{3}\pi$. ■

CAUTION! In using (4), we must pay attention to the quadrant in which z lies, since $\tan \theta$ has period π , so that the arguments of z and $-z$ have the same tangent. *Example:* for $\theta_1 = \arg(1 + i)$ and $\theta_2 = \arg(-1 - i)$ we have $\tan \theta_1 = \tan \theta_2 = 1$.

Argand Diagrams and Polar Form

This guide introduces Argand diagrams which are used to visualise complex numbers. It also shows how to calculate the modulus and argument of a complex number, their role in the polar form of a complex number and how to convert between Cartesian and polar forms.

Introduction

One of the problems that mathematicians came across when they were exploring the idea of complex numbers was their visualisation. A complex number does not appear anywhere on a real number line and so would not appear in coordinates in the xy -plane. This problem was solved in an ingenious way in the 19th century by Jean-Robert Argand (building on the work of Caspar Wessel). An **Argand diagram** uses the **real** and **imaginary parts** of a complex number as analogues of x and y in the Cartesian plane. The area of an Argand diagram is called the **complex plane** by mathematicians.

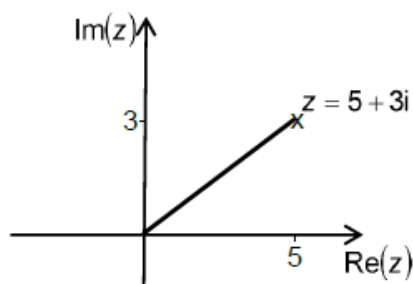
The study guide: [Basics of Complex Numbers](#) describes the **Cartesian form** of a complex number z as:

$$z = a + bi$$

Cartesian form of a complex number

Where a is the **real part** of a complex number, written $\text{Re}(z)$, and b is the **imaginary part** of a complex number, written $\text{Im}(z)$. In an Argand diagram the horizontal axis defines the real part of the complex number and the vertical axis defines the imaginary part. A complex number in Cartesian form has the coordinate (a, b) in an Argand diagram. A purely real number is positioned on the horizontal axis of an Argand diagram and a purely imaginary number is positioned on the vertical axis of an Argand diagram.

Example: Represent the complex number $z = 5 + 3i$ on an Argand diagram.



The position of the number $z = 5 + 3i$ is shown as an "x" but the number itself is usually represented as a line from the origin to the point.

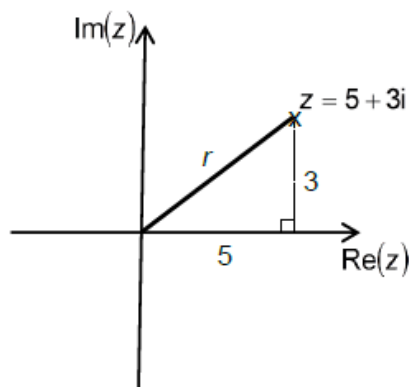
You should label the horizontal axis as $\text{Re}(z)$ and the vertical axis as $\text{Im}(z)$ in an Argand diagram.

The modulus and argument of a complex number

As soon as you represent a complex number visually you can begin to explore other ways of describing it. For example, an Argand diagram shows the similarity of a complex number with a vector, you can think of $\text{Re}(z)$ and $\text{Im}(z)$ being the number of steps along the real and imaginary axes respectively required to 'reach' z .

You can also get to z in a straight line directly from the origin. In order to do this all you need to know is what direction to face and how far to go. How far you need to go is the length of the line representing the complex number in an Argand diagram and is called the **modulus of the complex number**. It is very common to represent the modulus of a complex number z by the symbol $|z|$ or simply by the letter r . The value of the modulus is always positive.

You can use Pythagoras' theorem to find the modulus of a complex number (see study guide: [Pythagoras' Theorem](#)). Below, the Argand given for $z = 5 + 3i$ from page 1 of the guide has been annotated to reveal a right-angled triangle:



As you can see, the modulus of the complex number is the **hypotenuse** of the right-angled triangle. Using Pythagoras' theorem:

$$|z|^2 = r^2 = 5^2 + 3^2 = 34$$

so $|z| = r = \sqrt{34}$ (taking the positive root)

The modulus of a general complex number written in Cartesian form as $z = a + bi$ is:

$$|z| = r = \sqrt{a^2 + b^2}$$

Example: Find the modulus of the following complex numbers:

(a) $z = -2 - i$ (b) $z = -6i$ (c) $z = 3.4$

(a) Here $a = -2$ and $b = -1$ and so $|z| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{5}$

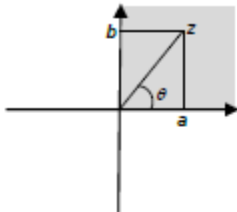
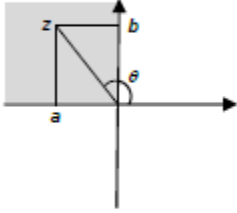
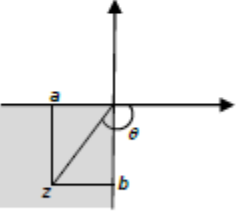
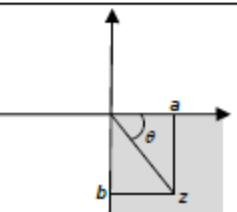
(b) Here $a = 0$ and $b = -6$ and so $|z| = \sqrt{0^2 + (-6)^2} = 6$

(c) Here $a = 3.4$ and $b = 0$ and so $|z| = \sqrt{3.4^2 + 0^2} = 3.4$

The modulus tells you how far to go from the origin to get to a complex number in an Argand diagram. You also need to know what direction the complex number is. This is known as the **argument** of a complex number which is written either as $\text{Arg}(z)$ or the Greek letter θ .

Arguments are **always measured in radians**
and take values in the range $-\pi < \theta \leq \pi$

Complex numbers either **on the real axis or in the upper half of the complex plane** have a **positive argument** measured anti-clockwise from the real axis. Complex numbers in the **lower half of the complex plane** have a **negative argument** measured clockwise from the real axis. You can use the inverse tangent (or arctan) function to find arguments. This can be confusing and the table below is designed to help you. It is a very good idea to sketch your complex number before trying to calculate its argument.

Quadrant (shaded)	Argand diagram sketch	Argument	
		Information	Formula
1		a is positive b is positive argument is positive	$\theta = \tan^{-1}\left(\frac{b}{a}\right)$
2		a is negative b is positive argument is positive	$\theta = \pi + \tan^{-1}\left(\frac{b}{a}\right)$
3		a is negative b is negative argument is negative	$\theta = -\pi + \tan^{-1}\left(\frac{b}{a}\right)$
4		a is positive b is negative argument is negative	$\theta = \tan^{-1}\left(\frac{b}{a}\right)$

The real axis is part of quadrants 1 and 2 (not quadrants 3 and 4). Given this:

Positive real numbers are in quadrant 1 and have an argument of 0.

Negative real numbers are in quadrant 2 and have an argument of π .

Also:

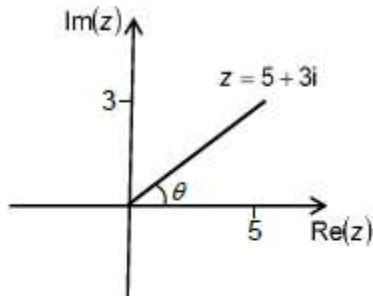
The argument of a purely imaginary number above the real axis (positive b) is $\pi/2$.

The argument of a purely imaginary number below the real axis (negative b) is $-\pi/2$.

Example: Find the modulus of the following complex numbers:

- (a) $z = 5 + 3i$ (b) $z = -2 - i$ (c) $z = -6i$ (d) $z = 3.4$

(a)



As you can see, $z = 5 + 3i$ is in quadrant 1.

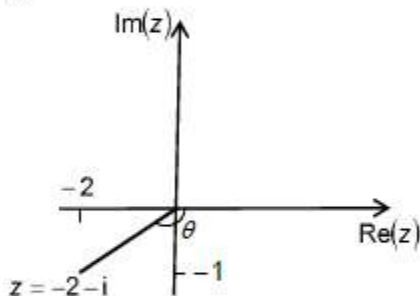
Also $\text{Re}(z) = a = 5$ and $\text{Im}(z) = b = 3$.

Using the appropriate formula from the table:

$$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{3}{5}\right) = 0.54 \text{ to 2 decimal places}$$

So for $z = 5 + 3i$, $\theta = \text{Arg}(z) = 0.54$ radians.

(b)



As you can see, $z = -2 - i$ is in quadrant 3.

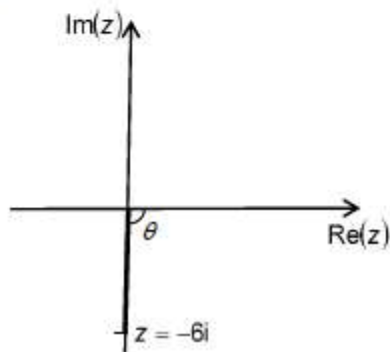
Also $\text{Re}(z) = a = -2$ and $\text{Im}(z) = b = -1$.

Using the appropriate formula from the table:

$$\theta = -\pi + \tan^{-1}\left(\frac{b}{a}\right) = -\pi + \tan^{-1}\left(\frac{-1}{-2}\right) = -2.68 \text{ to 2 d.p.}$$

So for $z = -2 - i$, $\theta = \text{Arg}(z) = -2.68$ radians.

(c)



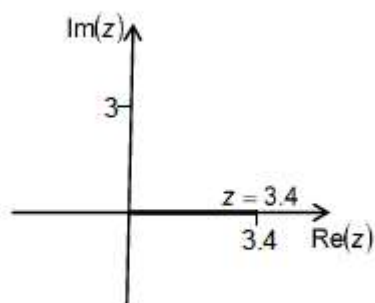
As you can see, $z = -6i$ is a purely imaginary

number below the real axis with $\text{Re}(z) = a = 0$ and

$\text{Im}(z) = b = -6$.

This means that $\theta = \text{Arg}(z) = -\pi/2$ as $z = -6i$ is at a right-angle to the real axis.

(d)



As you can see, $z = 3.4$ is a positive, purely real number with $\text{Re}(z) = a = 3.4$ and $\text{Im}(z) = b = 0$.

Positive, real numbers are defined as being in quadrant 1.

This means that $\theta = \text{Arg}(z) = 0$.

Polar form of a complex number

As you have seen, a complex number can be defined not only in terms of its real and imaginary parts but also in terms of its modulus and argument. Just as the Cartesian form of a complex number uses $\text{Re}(z)$ and $\text{Im}(z)$ there is another way of writing a complex number which uses $|z|$ and $\text{Arg}(z)$ called the polar form. By definition the polar form of a complex number is:

$$z = re^{i\theta}$$

where r is the modulus and θ is the argument of z

here e is the base of natural logarithm (see study guide: [Exponential Functions](#) for more details). The derivation of this form of a complex number is beyond the scope of this guide but a [Learning Enhancement Tutor](#) will be happy to explain it to you. The polar form is simply another way of writing a complex number, and you need to be comfortable converting from Cartesian form to polar form and vice-versa. The previous sections have shown you how to calculate the modulus and argument of complex number if you know the real and imaginary parts and so you can use those methods to help you convert a complex number from Cartesian form to polar form.

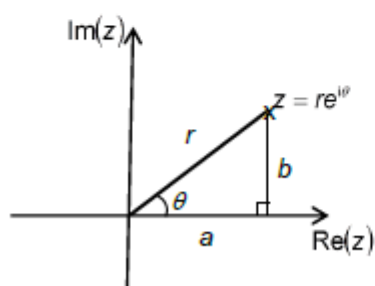
Example: Write the following complex numbers in polar form:

- (a) $z = 5 + 3i$ (b) $z = -2 - i$ (c) $z = -6i$ (d) $z = 3.4$

You can use the results from earlier in this guide to work out the polar form.

- (a) For $z = 5 + 3i$, $r = \sqrt{34}$ and $\theta = 0.54$ and so the polar form is $z = \sqrt{34}e^{0.54i}$.
- (b) For $z = -2 - i$, $r = \sqrt{5}$ and $\theta = -2.68$ and so the polar form is $z = \sqrt{5}e^{-2.68i}$.
- (c) For $z = -6i$, $r = 6$ and $\theta = -\pi/2$ and so the polar form is $z = 6e^{-\frac{\pi}{2}i}$.
- (d) For $z = 3.4$, $r = 3.4$ and $\theta = 0$ and so the polar form is $z = 3.4e^0 = 3.4$ as any number to the power 0 is equal to 1.

You now need to know how to convert a complex number in polar form to Cartesian form. To do this you also use trigonometry. Look at the Argand diagram below.



You can use trigonometric ratios to find a and b in terms of r and θ as you have a right-angled triangle. Here $\text{Re}(z) = a = r \cos \theta$ and $\text{Im}(z) = b = r \sin \theta$ and so you can write the Cartesian form of a complex number as:

$$z = r \cos \theta + r \sin \theta i$$

This is an important result that links complex numbers and trigonometry.

You can read the study guide: [Trigonometric Ratios: Sine, Cosine and Tangent](#) for more details about the definition of sine and cosine in a right-angled triangle.

Example: Write the following complex numbers in Cartesian form:

$$(a) \quad z = 2e^{1.2i} \quad (b) \quad z = \sqrt{3}e^{-1.4i} \quad (c) \quad z = e^{i\pi}$$

(a) For $z = 2e^{1.2i}$, $r = 2$ and $\theta = 1.2$ to convert to Cartesian form use:

$$\operatorname{Re}(z) = a = r \cos \theta: \quad \text{so (using radians)} \quad a = 2 \cos 1.2 = 0.72 \text{ to 2 decimal places.}$$

$$\operatorname{Im}(z) = b = r \sin \theta: \quad \text{so (using radians)} \quad b = 2 \sin 1.2 = 1.86 \text{ to 2 decimal places.}$$

So the Cartesian form of $z = 2e^{1.2i}$ is $z = 0.72 + 1.86i$.

(b) For $z = \sqrt{3}e^{-1.4i}$, $r = \sqrt{3}$ and $\theta = -1.4$ to convert to Cartesian form use:

$$\operatorname{Re}(z) = a = r \cos \theta: \quad \text{so (using radians)} \quad a = \sqrt{3} \cos(-1.4) = 0.29 \text{ to 2 decimal places.}$$

$$\operatorname{Im}(z) = b = r \sin \theta: \quad \text{so (using radians)} \quad b = \sqrt{3} \sin(-1.4) = -1.71 \text{ to 2 decimal places.}$$

So the Cartesian form of $z = \sqrt{3}e^{-1.4i}$ is $z = 0.29 - 1.71i$.

(c) For $z = e^{i\pi}$, $r = 1$ and $\theta = \pi$ to convert to Cartesian form use:

$$\operatorname{Re}(z) = a = r \cos \theta \quad \text{and so} \quad a = \cos \pi = -1.$$

$$\operatorname{Im}(z) = b = r \sin \theta \quad \text{and so} \quad b = \sin \pi = 0.$$

So the Cartesian form of $z = e^{i\pi}$ is $z = -1$. This is an important result and is discussed in the study guide: *Euler's Identity and De Moivre's Theorem*.

Triangle Inequality

Inequalities such as $x_1 < x_2$ make sense for *real* numbers, but not in complex because *there is no natural way of ordering complex numbers*. However, inequalities between absolute values (which are real), such as $|z_1| < |z_2|$ (meaning that z_1 is closer to the origin than z_2) are of great importance. The daily bread of the complex analyst is the **triangle inequality**

$$(6) \quad |z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{Fig. 326})$$

which we shall use quite frequently. This inequality follows by noting that the three points 0 , z_1 , and $z_1 + z_2$ are the vertices of a triangle (Fig. 326) with sides $|z_1|$, $|z_2|$, and $|z_1 + z_2|$, and one side cannot exceed the sum of the other two sides. A formal proof is left to the reader (Prob. 33). (The triangle degenerates if z_1 and z_2 lie on the same straight line through the origin.)

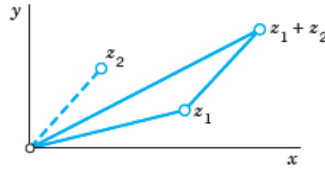


Fig. 326. Triangle inequality

By induction we obtain from (6) the **generalized triangle inequality**

$$(6^*) \quad |z_1 + z_2 + \cdots + z_n| \leq |z_1| + |z_2| + \cdots + |z_n|;$$

that is, *the absolute value of a sum cannot exceed the sum of the absolute values of the terms.*

EXAMPLE 2 Triangle Inequality

If $z_1 = 1 + i$ and $z_2 = -2 + 3i$, then (sketch a figure)

$$|z_1 + z_2| = |-1 + 4i| = \sqrt{17} = 4.123 < \sqrt{2} + \sqrt{13} = 5.020.$$

Multiplication and Division in Polar Form

This will give us a “geometrical” understanding of multiplication and division. Let

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1) \quad \text{and} \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2).$$

Multiplication. By (3) in Sec. 13.1 the product is at first

$$z_1 z_2 = r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)].$$

The addition rules for the sine and cosine (6) in App. A3.1 now yield

$$(7) \quad z_1 z_2 = r_1 r_2 \cos(\theta_1 + \theta_2) + i r_1 r_2 \sin(\theta_1 + \theta_2).$$

Taking absolute values on both sides of (7), we see that *the absolute value of a product equals the product of the absolute values of the factors*,

$$(8) \quad |z_1 z_2| = |z_1| |z_2|.$$

Taking arguments in (7) shows that *the argument of a product equals the sum of the arguments of the factors*,

$$(9) \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2 \quad (\text{up to multiples of } 2\pi).$$

Division. We have $z_1 = (z_1/z_2)z_2$. Hence $|z_1| = |(z_1/z_2)z_2| = |z_1/z_2||z_2|$ and by division by $|z_2|$

$$(10) \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (z_2 \neq 0).$$

Similarly, $\arg z_1 = \arg \left((z_1/z_2)z_2 \right) = \arg (z_1/z_2) + \arg z_2$ and by subtraction of $\arg z_2$

$$(11) \quad \arg \frac{z_1}{z_2} = \arg z_1 - \arg z_2 \quad (\text{up to multiples of } 2\pi).$$

Combining (10) and (11) we also have the analog of (7),

$$(12) \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} \cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2).$$

To comprehend this formula, note that it is the polar form of a complex number of absolute value r_1/r_2 and argument $\theta_1 - \theta_2$. But these are the absolute value and argument of z_1/z_2 , as we can see from (10), (11), and the polar forms of z_1 and z_2 .

EXAMPLE 3 Illustration of Formulas (8)–(11)

Let $z_1 = -2 + 2i$ and $z_2 = 3i$. Then $z_1 z_2 = -6 - 6i$, $z_1/z_2 = \frac{2}{3} + (\frac{2}{3})i$. Hence (make a sketch)

$$|z_1 z_2| = 6\sqrt{2} = 3\sqrt{8} = |z_1||z_2|, \quad |z_1/z_2| = 2\sqrt{2}/3 = |z_1|/|z_2|,$$

and for the arguments we obtain $\arg z_1 = 3\pi/4$, $\arg z_2 = \pi/2$,

$$\arg(z_1 z_2) = -\frac{3\pi}{4} = \arg z_1 + \arg z_2 - 2\pi, \quad \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{4} = \arg z_1 - \arg z_2. \quad \blacksquare$$

EXAMPLE 4 Integer Powers of z . De Moivre's Formula

From (8) and (9) with $z_1 = z_2 = z$ we obtain by induction for $n = 0, 1, 2, \dots$

$$(13) \quad z^n = r^n (\cos n\theta + i \sin n\theta).$$

Similarly, (12) with $z_1 = 1$ and $z_2 = z^n$ gives (13) for $n = -1, -2, \dots$. For $|z| = r = 1$, formula (13) becomes **De Moivre's formula**³

$$(13^*) \quad (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

We can use this to express $\cos n\theta$ and $\sin n\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$. For instance, for $n = 2$ we have on the left $\cos^2 \theta + 2i \cos \theta \sin \theta - \sin^2 \theta$. Taking the real and imaginary parts on both sides of (13) with $n = 2$ gives the familiar formulas

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \cos \theta \sin \theta.$$

This shows that *complex* methods often simplify the derivation of *real* formulas. Try $n = 3$. ■

Roots

If $z = w^n$ ($n = 1, 2, \dots$), then to each value of w there corresponds *one* value of z . We shall immediately see that, conversely, to a given $z \neq 0$ there correspond precisely n distinct values of w . Each of these values is called an **n th root** of z , and we write

$$(14) \quad w = \sqrt[n]{z}.$$

Hence this symbol is ***multivalued***, namely, *n -valued*. The n values of $\sqrt[n]{z}$ can be obtained as follows. We write z and w in polar form

$$z = r(\cos \theta + i \sin \theta) \quad \text{and} \quad w = \mathcal{R}(\cos \phi + i \sin \phi).$$

Then the equation $w^n = z$ becomes, by De Moivre's formula (with ϕ instead of θ),

$$w^n = \mathcal{R}^n (\cos n\phi + i \sin n\phi) = z = r(\cos \theta + i \sin \theta).$$

The absolute values on both sides must be equal; thus, $\mathcal{R}^n = r$, so that $\mathcal{R} = \sqrt[n]{r}$, where $\sqrt[n]{r}$ is positive real (an absolute value must be nonnegative) and thus uniquely determined. Equating the arguments $n\phi$ and θ and recalling that θ is determined only up to integer multiples of 2π , we obtain

$$n\phi = \theta + 2k\pi, \quad \text{thus} \quad \phi = \frac{\theta}{n} + \frac{2k\pi}{n}$$

where k is an integer. For $k = 0, 1, \dots, n-1$ we get n *distinct* values of w . Further integers of k would give values already obtained. For instance, $k = n$ gives $2k\pi/n = 2\pi$, hence the w corresponding to $k = 0$, etc. Consequently, $\sqrt[n]{z}$, for $z \neq 0$, has the n distinct values

$$(15) \quad \sqrt[n]{z} = \sqrt[n]{r} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right)$$

where $k = 0, 1, \dots, n-1$. These n values lie on a circle of radius $\sqrt[n]{r}$ with center at the origin and constitute the vertices of a regular polygon of n sides. The value of $\sqrt[n]{z}$ obtained by taking the principal value of $\arg z$ and $k = 0$ in (15) is called the **principal value** of $w = \sqrt[n]{z}$.

Taking $z = 1$ in (15), we have $|z| = r = 1$ and $\text{Arg } z = 0$. Then (15) gives

$$(16) \quad \sqrt[n]{1} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad k = 0, 1, \dots, n-1.$$

These n values are called the **n th roots of unity**. They lie on the circle of radius 1 and center 0, briefly called the **unit circle** (and used quite frequently). Figures 327–329 show $\sqrt[3]{1} = 1, -\frac{1}{2} \pm \frac{1}{2}\sqrt{3}i$, $\sqrt[4]{1} = \pm 1, \pm i$, and $\sqrt[5]{1}$.

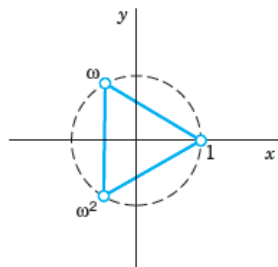


Fig. 327. $\sqrt[3]{1}$

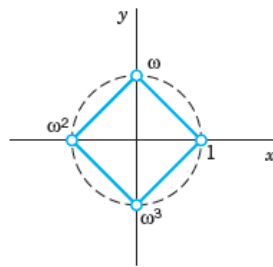


Fig. 328. $\sqrt[4]{1}$

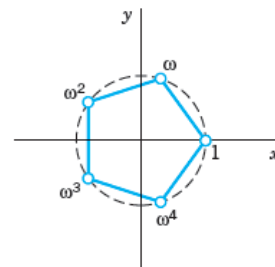


Fig. 329. $\sqrt[5]{1}$

If ω denotes the value corresponding to $k = 1$ in (16), then the n values of $\sqrt[n]{1}$ can be written as

$$1, \omega, \omega^2, \dots, \omega^{n-1}.$$

More generally, if w_1 is any n th root of an arbitrary complex number $z (\neq 0)$, then the n values of $\sqrt[n]{z}$ in (15) are

$$(17) \quad w_1, \quad w_1\omega, \quad w_1\omega^2, \quad \dots, \quad w_1\omega^{n-1}$$

because multiplying w_1 by ω^k corresponds to increasing the argument of w_1 by $2k\pi/n$. Formula (17) motivates the introduction of roots of unity and shows their usefulness.

Recommended Ex: 13.1 and 13.2

Book: Advanced Engineering Mathematics

By Erwin Kreyszig